Transkriptomanalyse

12. November 2026, Sommersemester 2025

Aufgabe 3

a)

(i)

z.Z.: $T(\underline{x}) = \sum_{n=1}^{N} \sum_{n'=1}^{N} \|\underline{x}_n - \underline{x}_{n'}\|^2 = 2N \sum_{n=1}^{N} \|\underline{x}_n - \overline{\underline{x}}\|^2$

$$T(\underline{x}) = \sum_{n=1}^{N} \sum_{n'=1}^{N} \|\underline{x}_n - \underline{x}_{n'}\|^2$$

$\Rightarrow$ gleichmäßige Verschiebung $\rightarrow$ Abstand bleibt gleich

$$T(\underline{x}) = \sum_{n=1}^{N} \sum_{n'=1}^{N} \|(\underline{x}_n - \overline{\underline{x}}) - (\underline{x}_{n'} - \overline{\underline{x}})\|^2$$

$$= \sum_{n=1}^{N} \sum_{n'=1}^{N} \left(\|(\underline{x}_n - \overline{\underline{x}})\|^2 + \|(\underline{x}_{n'} - \overline{\underline{x}})\|^2 - 2\langle (\underline{x}_n - \overline{\underline{x}}), (\underline{x}_{n'} - \overline{\underline{x}}) \rangle \right) \quad | \text{ binom. Formel}$$

$$= \sum_{n=1}^{N} \sum_{n'=1}^{N} \|(\underline{x}_n - \overline{\underline{x}})\|^2 + \sum_{n=1}^{N} \sum_{n'=1}^{N} \|(\underline{x}_{n'} - \overline{\underline{x}})\|^2 - \sum_{n=1}^{N} \sum_{n'=1}^{N} 2\langle (\underline{x}_n - \overline{\underline{x}}), (\underline{x}_{n'} - \overline{\underline{x}}) \rangle$$

$$= N \sum_{n=1}^{N} \|(\underline{x}_n - \overline{\underline{x}})\|^2 + N \sum_{n'=1}^{N} \|(\underline{x}_{n'} - \overline{\underline{x}})\|^2 - \sum_{n=1}^{N} \sum_{n'=1}^{N} 2\langle (\underline{x}_n - \overline{\underline{x}}), (\underline{x}_{n'} - \overline{\underline{x}}) \rangle$$

$$= 2N \sum_{n=1}^{N} \|(\underline{x}_n - \overline{\underline{x}})\|^2 - \sum_{n=1}^{N} \sum_{n'=1}^{N} 2\langle (\underline{x}_n - \overline{\underline{x}}), (\underline{x}_{n'} - \overline{\underline{x}}) \rangle$$

$\Rightarrow$ Biliniarität von Skalarprodukten

$$T(\underline{x}) = 2N \sum_{n=1}^{N} \|(\underline{x}_n - \overline{\underline{x}})\|^2 - 2 \left\langle \sum_{n=1}^{N} (\underline{x}_n - \overline{\underline{x}}), \sum_{n'=1}^{N} (\underline{x}_{n'} - \overline{\underline{x}}) \right\rangle$$

Nebenrechnung:

$$\begin{aligned}\sum_{n=1}^{N}(\underline{x}_n - \overline{\underline{x}}) &= \sum_{n=1}^{N}\underline{x}_n - \sum_{n=1}^{N}\overline{\underline{x}} \\ &= \sum_{n=1}^{N}\underline{x}_n - N\cdot\overline{\underline{x}} \\ &= \sum_{n=1}^{N}\underline{x}_n - N\cdot\frac{1}{N}\sum_{n=1}^{N}\underline{x}_n \\ &= 0\end{aligned}$$

Ergebnis nutzen:

$$\begin{aligned}T(\underline{\underline{x}}) &= 2N\sum_{n=1}^{N}\|(\underline{x}_n - \overline{\underline{x}})\|^2 - 2\left\langle\sum_{n=1}^{N}(\underline{x}_n - \overline{\underline{x}}), \sum_{n'=1}^{N}(\underline{x}_{n'} - \overline{\underline{x}})\right\rangle \\ &= 2N\sum_{n=1}^{N}\|(\underline{x}_n - \overline{\underline{x}})\|^2 - 2\langle 0,0\rangle \\ &= 2N\sum_{n=1}^{N}\|(\underline{x}_n - \overline{\underline{x}})\|^2\end{aligned}$$

(ii)

z.Z.: $T(\underline{\underline{x}}) = 2N\sum_{n=1}^{N}\|\underline{x}_n - \overline{\underline{x}}\|^2 = 2N^2\sum_{l=1}^{L}Var(\underline{x}_l)$

$$\begin{aligned}T(\underline{\underline{x}}) &= 2N\sum_{n=1}^{N}\|\underline{x}_n - \overline{\underline{x}}\|^2 \\ &= 2N\sum_{n=1}^{N}\sum_{l=1}^{L}(x_{nl} - \overline{x}_l)^2 && \text{| Def. quadratische euklidische Distanz} \\ &= 2N\left(\sum_{n=1}^{N}\sum_{l=1}^{L}(x_{nl} - \overline{x}_l)^2\right)\cdot\frac{N}{N} \\ &= 2N^2\sum_{l=1}^{L}\left(\frac{1}{N}\cdot\sum_{n=1}^{N}(x_{nl} - \overline{x}_l)^2\right) \\ &= 2N^2\sum_{l=1}^{L}Var(\underline{x}_l) && \text{| Def. empirische Varianz}\end{aligned}$$

(iii)

z.Z.: $W(\underline{\underline{x}}, \underline{u}) = \sum_{n=1}^{N}\sum_{n'=1}^{N}\|\underline{x}_n - \underline{x}_{n'}\|^2\delta_{u_n,u_{n'}} = 2\sum_{k=1}^{K}N_k\sum_{n=1}^{N}\|\underline{x}_n - \overline{\underline{x}}_k\|^2\delta_{u_n,k}$

$$
\begin{aligned}
W(\underline{\underline{x}}, \underline{\underline{u}}) &= \sum_{n=1}^{N} \sum_{n'=1}^{N} \|\underline{x}_n - \underline{x}_{n'}\|^2 \delta_{u_n, u_{n'}} \\
&\Rightarrow \delta_{u_n, u_{n'}} = \sum_{k=1}^{K} \delta_{u_n,k} \delta_{u_{n'},k} \\
W(\underline{\underline{x}}, \underline{\underline{u}}) &= \sum_{k=1}^{K} \sum_{n=1}^{N} \sum_{n'=1}^{N} \|\underline{x}_n - \underline{x}_{n'}\|^2 \delta_{u_n,k} \delta_{u_{n'},k} \\
&= \sum_{k=1}^{K} \sum_{n=1}^{N} \sum_{n'=1}^{N} \left(\|(\underline{x}_n - \overline{\underline{x}}_k)\|^2 + \|(\underline{x}_{n'} - \overline{\underline{x}}_k)\|^2 - 2\langle (\underline{x}_n - \overline{\underline{x}}_k), (\underline{x}_{n'} - \overline{\underline{x}}_k) \rangle \right) \delta_{u_n,k} \delta_{u_{n'},k} \\
&= \sum_{k=1}^{K} \Big(\sum_{n=1}^{N} \sum_{n'=1}^{N} \|(\underline{x}_n - \overline{\underline{x}}_k)\|^2 \delta_{u_n,k} \delta_{u_{n'},k} + \sum_{n=1}^{N} \sum_{n'=1}^{N} \|(\underline{x}_{n'} - \overline{\underline{x}}_k)\|^2 \delta_{u_n,k} \delta_{u_{n'},k} \\
&\quad - \sum_{n=1}^{N} \sum_{n'=1}^{N} 2\langle (\underline{x}_n - \overline{\underline{x}}_k), (\underline{x}_{n'} - \overline{\underline{x}}_k) \rangle \delta_{u_n,k} \delta_{u_{n'},k} \Big)
\end{aligned}
$$

Nebenrechnung:

$$
\sum_{n=1}^{N} \sum_{n'=1}^{N} 2\langle (\underline{x}_n - \overline{\underline{x}}_k), (\underline{x}_{n'} - \overline{\underline{x}}_k) \rangle \delta_{u_n,k} \delta_{u_{n'},k} = 2 \left\langle \sum_{n=1}^{N} (\underline{x}_n - \overline{\underline{x}}_k) \delta_{u_n,k}, \sum_{n'=1}^{N} (\underline{x}_{n'} - \overline{\underline{x}}_k) \delta_{u_{n'},k} \right\rangle
$$

$$
\begin{aligned}
\sum_{n=1}^{N} (\underline{x}_n - \overline{\underline{x}}_k) \delta_{u_n,k} &= \sum_{n \in C_k} (\underline{x}_n - \overline{\underline{x}}_k) \\
&= \sum_{n \in C_k} \underline{x}_n - N_k \cdot \overline{\underline{x}}_k \\
&= \sum_{n \in C_k} \underline{x}_n - N_k \cdot \frac{1}{N_k} \sum_{n \in C_k} \underline{x}_n = 0 \\
\Rightarrow \sum_{n=1}^{N} \sum_{n'=1}^{N} 2\langle (\underline{x}_n - \overline{\underline{x}}_k), (\underline{x}_{n'} - \overline{\underline{x}}_k) \rangle \delta_{u_n,k} \delta_{u_{n'},k} &= 0
\end{aligned}
$$

Ergebnis nutzen:

$$
\begin{aligned}
W(\underline{\underline{x}}, \underline{\underline{u}}) &= \sum_{k=1}^{K} \left(\sum_{n=1}^{N} \sum_{n'=1}^{N} \|(\underline{x}_n - \overline{\underline{x}}_k)\|^2 \delta_{u_n,k} \delta_{u_{n'},k} + \sum_{n=1}^{N} \sum_{n'=1}^{N} \|(\underline{x}_{n'} - \overline{\underline{x}}_k)\|^2 \delta_{u_n,k} \delta_{u_{n'},k} - 0 \right) \\
&= \sum_{k=1}^{K} \left(\sum_{n=1}^{N} \|(\underline{x}_n - \overline{\underline{x}}_k)\|^2 \delta_{u_n,k} \sum_{n'=1}^{N} \delta_{u_{n'},k} + \sum_{n'=1}^{N} \|(\underline{x}_{n'} - \overline{\underline{x}}_k)\|^2 \delta_{u_{n'},k} \sum_{n=1}^{N} \delta_{u_n,k} \right) \\
&= \sum_{k=1}^{K} \left(\sum_{n=1}^{N} \|(\underline{x}_n - \overline{\underline{x}}_k)\|^2 \delta_{u_n,k} \cdot N_k + \sum_{n'=1}^{N} \|(\underline{x}_{n'} - \overline{\underline{x}}_k)\|^2 \delta_{u_{n'},k} \cdot N_k \right) \\
&= \sum_{k=1}^{K} \left(2N_k \sum_{n=1}^{N} \|(\underline{x}_n - \overline{\underline{x}}_k)\|^2 \delta_{u_n,k} \right)
\end{aligned}
$$

(iv)

z.Z.: $W(\underline{\underline{x}}, \underline{u}) = 2\sum_{k=1}^{K} N_k \sum_{n=1}^{N} \|\underline{x}_n - \overline{\underline{x}}_k\|^2 \delta_{u_n,k} = 2\sum_{k=1}^{K} N_k^2 \sum_{l=1}^{L} Var(\underline{x}_l|k)$

$$
\begin{aligned}
W(\underline{\underline{x}}, \underline{u}) &= 2\sum_{k=1}^{K} N_k \sum_{n=1}^{N} \|\underline{x}_n - \overline{\underline{x}}_k\|^2 \delta_{u_n,k} \\
&= 2\sum_{k=1}^{K} N_k \sum_{n=1}^{N} \sum_{l=1}^{L} (x_{nl} - \overline{x}_{kl})^2 \delta_{u_n,k} && \text{| Def. quadratische euklidische Distanz} \\
&= 2\sum_{k=1}^{K} N_k \sum_{l=1}^{L} \sum_{n=1}^{N} (x_{nl} - \overline{x}_{kl})^2 \delta_{u_n,k} \\
&= 2\sum_{k=1}^{K} N_k \sum_{l=1}^{L} \sum_{n \in C_k} (x_{nl} - \overline{x}_{kl})^2 && \text{| betrachte nur die } n \text{ die zu Cluster } C_k \text{ gehören} \\
&= 2\sum_{k=1}^{K} N_k \sum_{l=1}^{L} \sum_{n \in C_k} (x_{nl} - \overline{x}_{kl})^2 \cdot \frac{N_k}{N_k} \\
&= 2\sum_{k=1}^{K} N_k^2 \sum_{l=1}^{L} \left(\frac{1}{N_k} \sum_{n \in C_k} (x_{nl} - \overline{x}_{kl})^2 \right) \\
&= 2\sum_{k=1}^{K} N_k^2 \sum_{l=1}^{L} Var(\underline{x}_l|k) && \text{| Def. empirische Varianz gegeben: Zugehörigkeit } k
\end{aligned}
$$

b)

z.Z.: $\underline{\mu}^* = \arg\min_{\underline{\mu}} \sum_{n=1}^{N} \|\underline{x}_n - \underline{\mu}\|^2 = \frac{1}{N} \sum_{n=1}^{N} \underline{x}_n = \overline{\underline{x}}$

$$
\begin{aligned}
f(\underline{\mu}) &= \sum_{n=1}^{N} \|\underline{x}_n - \underline{\mu}\|^2 \\
&= \sum_{n=1}^{N} \sum_{l=1}^{L} (x_{nl} - \mu_l)^2 \\
&= \sum_{l=1}^{L} \sum_{n=1}^{N} (x_{nl} - \mu_l)^2
\end{aligned}
$$

Jede Komponente μ_l in $\underline{\mu}$ kann einzeln betrachtet und minimiert werden, da sie separierbar sind:

Ableitung:

$$\begin{aligned}\frac{\partial f}{\partial \mu_l} &= \sum_{n=1}^{N} \frac{\partial}{\partial \mu_l}(x_{nl} - \mu_l)^2 \\ &= \sum_{n=1}^{N} -2(x_{nl} - \mu_l)\end{aligned}$$

Null setzen:

$$\begin{aligned}0 &= \sum_{n=1}^{N} -2(x_{nl} - \mu_l) \\ &= -2\left(\sum_{n=1}^{N} x_{nl} - N\mu_l\right) \\ \Leftrightarrow \mu_l &= \frac{1}{N}\sum_{n=1}^{N} x_{nl}\end{aligned}$$

Daher:

$$\underline{\mu}^* = (\mu_1, \ldots, \mu_L)^T = \frac{1}{N}\sum_{n=1}^{N} \underline{x}_n$$